D-BLOCK

(1) Zn, Cd, Hg ← transition ✗✗ but d-block ✓

(2) $Cr \rightarrow 3d^5 4s^1$, $Cu \rightarrow 3d^{10} 4s^1$, $Nb \rightarrow 5s^1 4d^4$,

$Mo \rightarrow 5s^1 4d^5$

$Tc \rightarrow 5s^1 4d^6$

$Ru \rightarrow 5s^1 4d^7$

$Rh \rightarrow 5s^1 4d^8$

$Pd \rightarrow 5s^0 4d^{10}$

$Ag \rightarrow 5s^1 4d^{10}$

$\rightarrow s^2 d^n x$

$s^1 d^{n+1}$ ✓

$Pt \rightarrow 6s^1 5d^9$,

$Au \rightarrow 6s^1 5d^{10}$

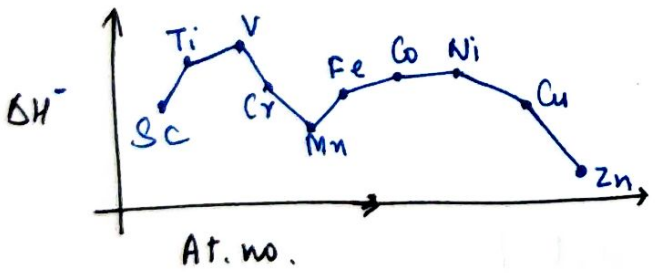
EXCEPTIONS

(3) i) Varia. O.S , ii) Coloured ions , iii) Complex for. , iv) cataly. Prop.
v) Paramag. behaviour

(4) Zn, Cd, Hg, Mn → hcp, bcc jaisa nhi hai , Hg → Liq.

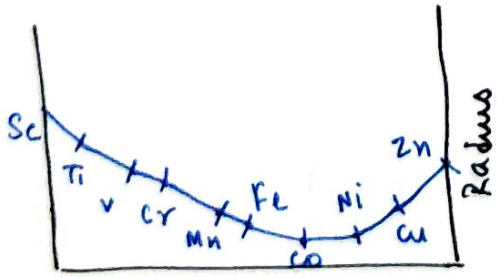
(5) M.p middle tak ↑ then decrease . Exception Mn and Tc.
W ka most M.P ↑ . Cr | Mo | W apni row mein most MP.

(6)



V most ΔH⁻

(7)



(8) Meta. Radii ↓ + At. mass ↑ = Density generally ↑

(9) I.E → i) Removal of one e⁻ alters the relative energies of 4s and 3d orbitals.

ii) Doubly or highly charged ions have d^n confi. and no 4s e⁻s.

3 terms for IE → i) att. of e⁻ by nucleus

ii) rep. b/t e⁻

iii) **exchange energy** → ↺ { total no. of possi. pairs of parallel spins in degenerate orbitals }

★ No loss of exchange energy at d^6 confi

(10)

Element	Most Common O.S
[Sc	+3 (ek he O.S hai)]
Ti	+4
V	+5
[Cr	+3, +6]
[Mn	+2, +7] → Most no. of O.states of Mn
Fe	+2, +3
Co	+2, +3
Ni	+2
[Cu	+2]
[Zn	+2 (ek he O.S state]

[ ] → imp.

(11) $Ti^{+4} > Ti^{+3}, Ti^{+2}$ (Stabi.)

(12) P-Block → low O.S favoured by heavier members

D-block → Higher O.S " " lighter "

e.g → $Mo^{+6}, W^{+6} > Cr^{+6}$ (Stabi.)

∴ MoO_3, WO_3 → not O.A but $Cr_2O_7^{2-}$ is O.A

(13) Low O.S when ligands are π e⁻ acceptor with σ bond (SYNERGIC BONDING)

(14) Only oxidising acids react with Cu

SRP for [Mn, Zn, Ni] are more -ve than expected.

(15) V^{+2} is half filled t_{2g} stable

(16) i) [TiX_4, VF_5, CrF_6] → highest o.state

ii) [MnO_3F] ✗ Known, [FeX_3, CoF_3] Known

iii) [VX_2] unstable

iv) [VOX_3] obtained by hydrolysis of halides

(17) i) $(FeO_4)^{2-} \xrightarrow{\text{alkali}} Fe_2O_3 + O_2$

ii) MnF_4 highest Mn fluoride [as 'O' can form multiple bonds]

iii) Mn_2O_7 " oxide

(18) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ (Oxidis. Power)

(19) i) Ti, V → Passive to non oxidis. acids

ii) Ni → highest -ve enthalpy of hydration

iii) Mn^{+3}, Co^{+3} → Strong O.A

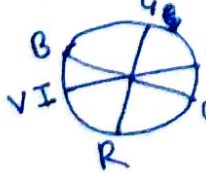
iv) Ti^{+2}, V^{2+}, Cr^{+2} → Strong R.A

(20) [$\mu = \sqrt{n(n+2)}$] (SPIN ONLY FORMULA) (Unit → BM (Bohr Magneton)

↳ Sc to Zn mein obs. value generally ↑ w.r.t calculated

(21) i) Colour due to d-d transi.

ii) Complementary colour obs.



] → Jiske opp. hai wohi Complementary colour hoga

e.g → Blue absorbed toh orange aaega

★ Fe^{+3} → yellow, Fe^{+2} → green

(22) Forms Complex Comp. due to →

i) small size of metal ions , ii) high ionic charges
iii) avail. of d-orb. for bond forma.

(23) Catalysts bcz → i) adopt multiple O.S
ii) form complex

(24) Interstitial Comp. → i) nor ionic nor covalent
ii) M.P ↑
iii) Hard , Borides ≈ Diamonds (hardness)
iv) Conductivity good
v) chemically inert

(25) Brass → Cu + Zn , Bronze → Cu, Sn

(26) O.S no. ↑ , ionic chara. ↓ ,
V_2O_5 → amphoteric

(27) $CrO < Cr_2O_3 < CrO_3$
(Bas.) (Ampho.) (Acidic)

$V_2O_3 < V_2O_4 < V_2O_5$
(Bas.) (Less Bas.) (Ampho.)

(28) $V_2O_5 \xrightarrow[\text{OR}]{\text{alkali}} VO_4^{3-} + VO_4^{+}$
acids

SOME IMP. POINTS

① $Zn > Cu$ (IE_1), $Zn < Cu$ (IE_2)

② $Cr > Mn$ (IE_2) , $Mn > Cr$ (IE_3)

$d^5\ 4s^2 \xrightarrow{-1e^-} d^6$

③ IE → discontinuity → alters rel. energies of 4s, 3d orb. d orbi. ki energy ↓↓ ki all e^- go to d-orbi.

④ $E^\circ_{Cu^{2+}/Cu} = +0.34V$ & bcz. $\boxed{\Delta H^\circ_{atomi.} + IE_1 + IE_2}$ not balanced by $\Delta H_{hyd.}$

⑤ $Cr^{+2} \xrightarrow{-1e^-} Cr^{+3}$ $(+0.41V)$

↓

d^3 half filled → t_{2g} half filled stable

⑥ $Mn^{+3} \xrightarrow{+e^-} Mn^{+2}$ $(+1.57V$

(d^5)

⑦ $Ni^{+2}/Ni = -0.25V$ (Zyada -ve as more -ve HE)

⑧ $\boxed{Fe^{+3}/Fe^{+2} = +0.77V}$

⑨ MnO_4^-, $Cr_2O_7^{2-}$, CrO_4^{2-}] → Coloured due to charge transfer phenomena

⑩ $2Cu^{+1} \longrightarrow Cu^{+2} + Cu$

$Cu^{+2} > Cu^{+1}$ (aq.) as more ~~for~~ HE for Cu^{+2} (Stabi.)

⑪ Zn/Cd/Hg → MPL, BPL as → [only d e⁻s not suff., they should be unpaired (1 unpaired e⁻ per d-orbital)]

⑫ $FeCl_3 \xrightarrow[673K]{\text{dimerise}} Fe_2Cl_6$ (3C-4e⁻)

(13) $ns e^- + (n-1)d e^-$ paarti. in bond. , ∴ M.P, B.P ↑↑ of Transi. elements

(14) $MnO < Mn_2O_3 < MnO_2 < Mn_2O_7$
(Basic) (Less Bas) (Amphs) (Acidic)

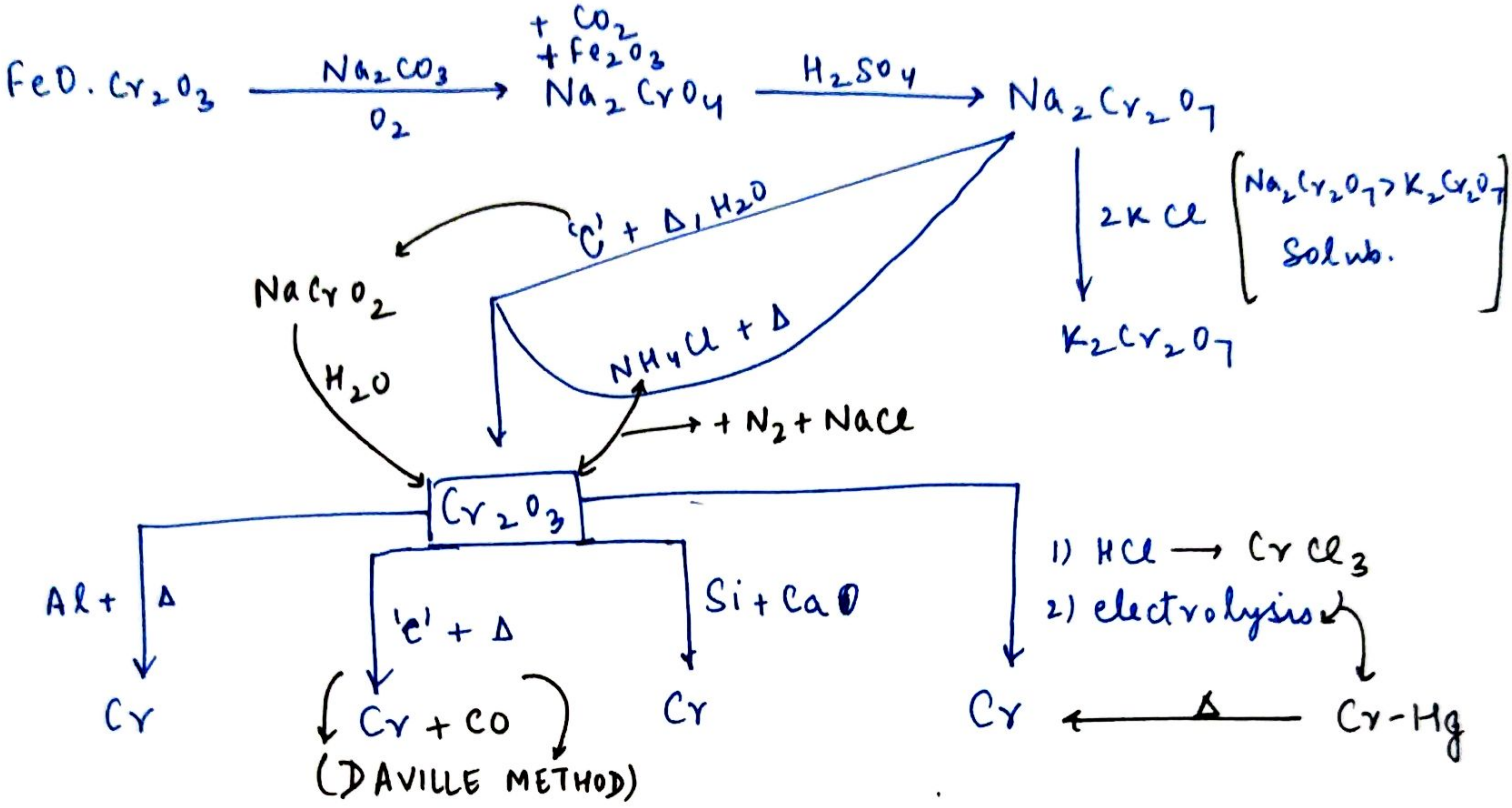
(15) $Fe^{+3} + I^- \longrightarrow I_2 + Fe^{+2}$

$Fe^{+2} + S_2O_8^{2-} \longrightarrow SO_4^{-2} + Fe^{+3}$

[Fe^{+3} catalyses $I^- + S_2O_8^{2-}$ reacn]

ORES →

I. Chromite Ore ($FeO.Cr_2O_3$)



① yellow - orange eqb. →

$$2H^+ + 2CrO_4^{2-} \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} Cr_2O_7^{2-} + H_2O$$

(yellow) (orangle)

② CO → burns with blue edged flame

③ $Cr_2O_3 + Si \longrightarrow SiO_2 + Cr$

$CaO + SiO_2 \longrightarrow CaSiO_3$

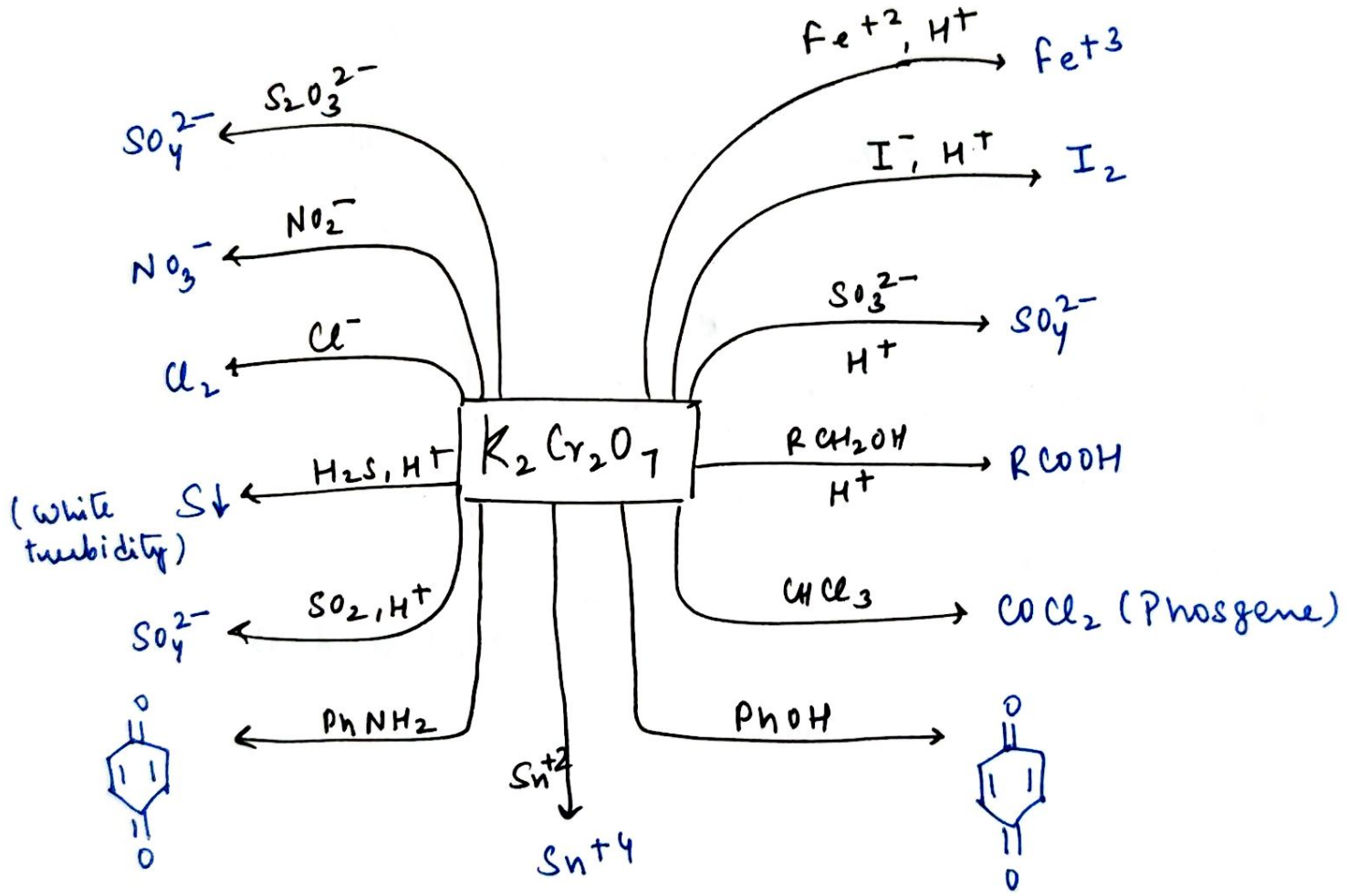
$K_2Cr_2O_7$

$\xrightarrow{\Delta}$ $K_2CrO_4 + Cr_2O_3 + O_2$

$Cr_2O_7^{2-}$ → O.Agent

$Cr_2O_7^{2-} \xrightarrow{H^+} Cr^{+3} + H_2O$ (+1.33 V)

★

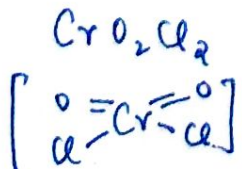


★ CHROMYL CHLORIDE TEST

Cl^- salt $\xrightarrow[H_2SO_4 \text{ (conc)}]{K_2Cr_2O_7}$ CrO_2Cl_2 [Deep Reddish orange ↑] $\xrightarrow{NaOH}$ $Na_2CrO_4 + NaCl$ (yellow)

$Na_2CrO_4 \xrightarrow{Pb(CH_3COO)_2}$ $PbCrO_4\downarrow$ (yellow)

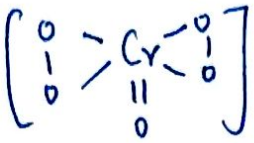
$CrO_2Cl_2 \xrightarrow{H_2O} H_2CrO_4 + 2HCl$



$K_2Cr_2O_7 + H_2O_2$ $\xrightarrow{\text{ether}}$ CrO_5 (Deep blue)

$+H_2SO_4$

$K_2SO_4 + CrO_5 + H_2O$



Blue fades away

$Cr^{+3} + O_2$

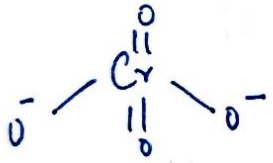
$K_2Cr_2O_7 \xrightarrow[H_2SO_4]{\text{cold, conc.}} CrO_3 + K_2SO_4 + H_2O$

(Red needle like crystals)

$K_2Cr_2O_7 \xrightarrow[\text{warm}]{H_2SO_4} K_2SO_4 + Cr_2(SO_4)_3 + \boxed{3[O]} + H_2O$

O.A

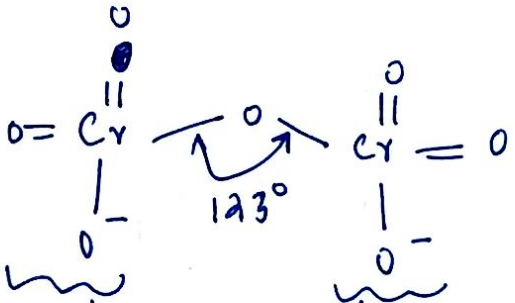
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$\boxed{CrO_4^{2-}}$

Tetrahedral

,

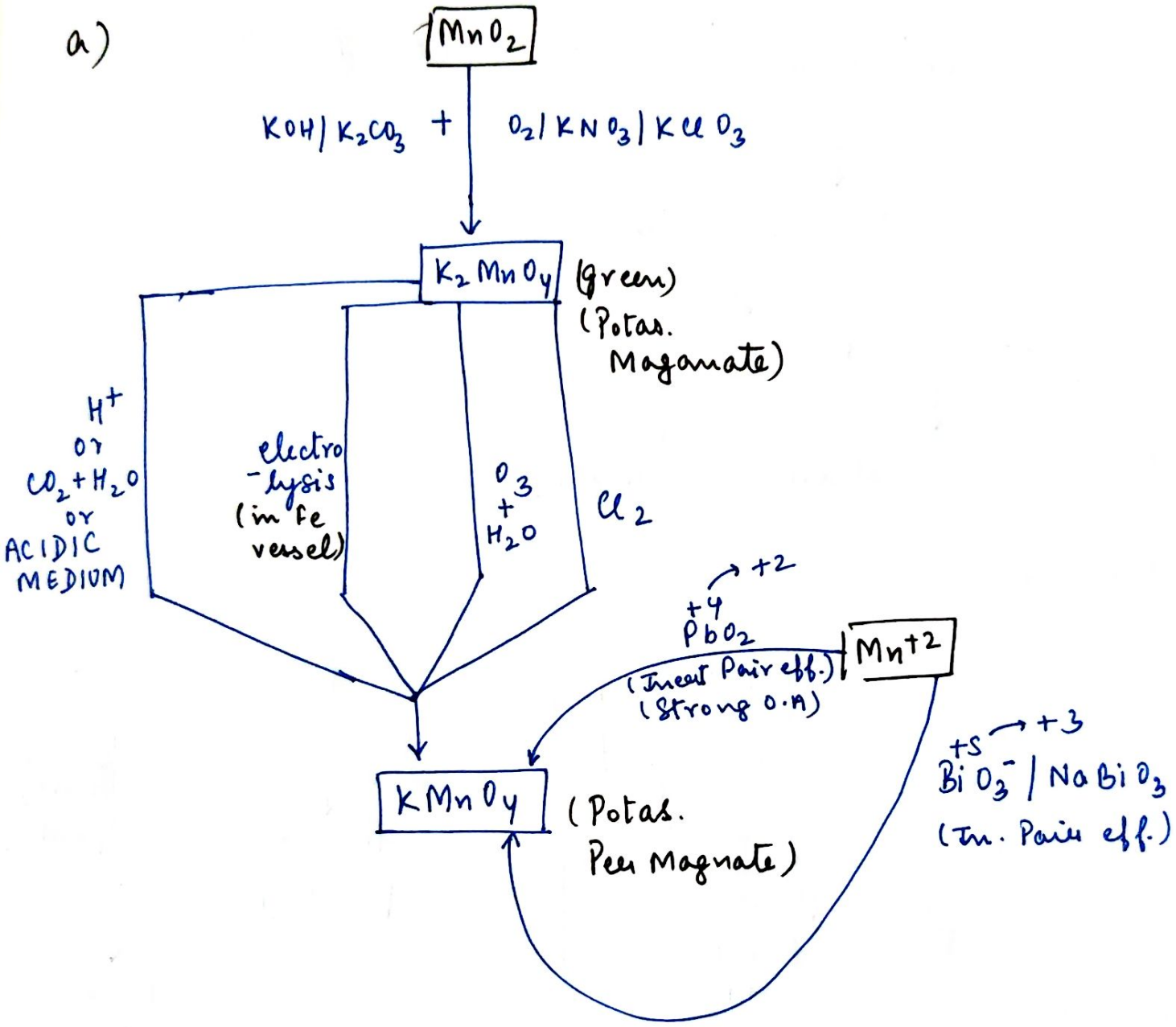


123°

Tetrahedral

II PYROLUSITE (MnO_2)

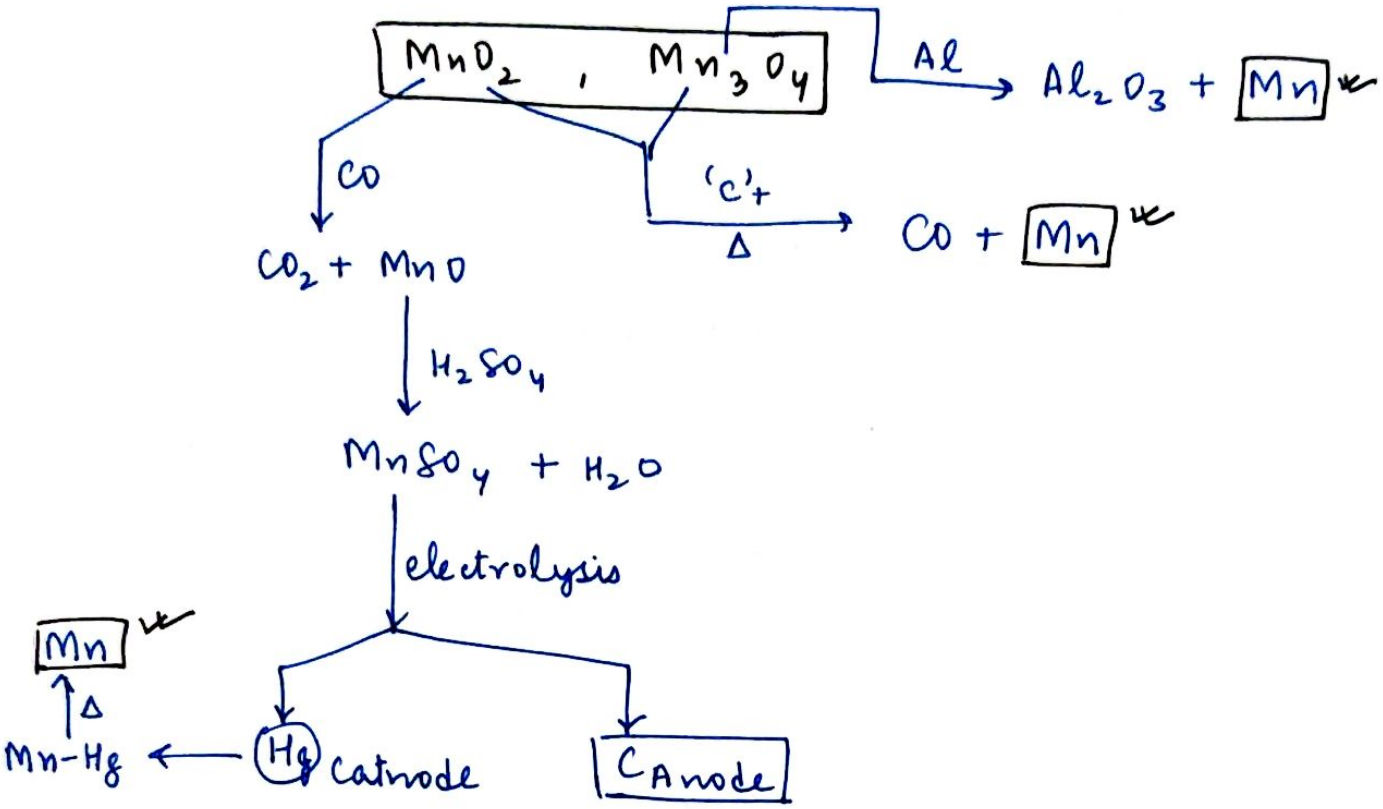
a)



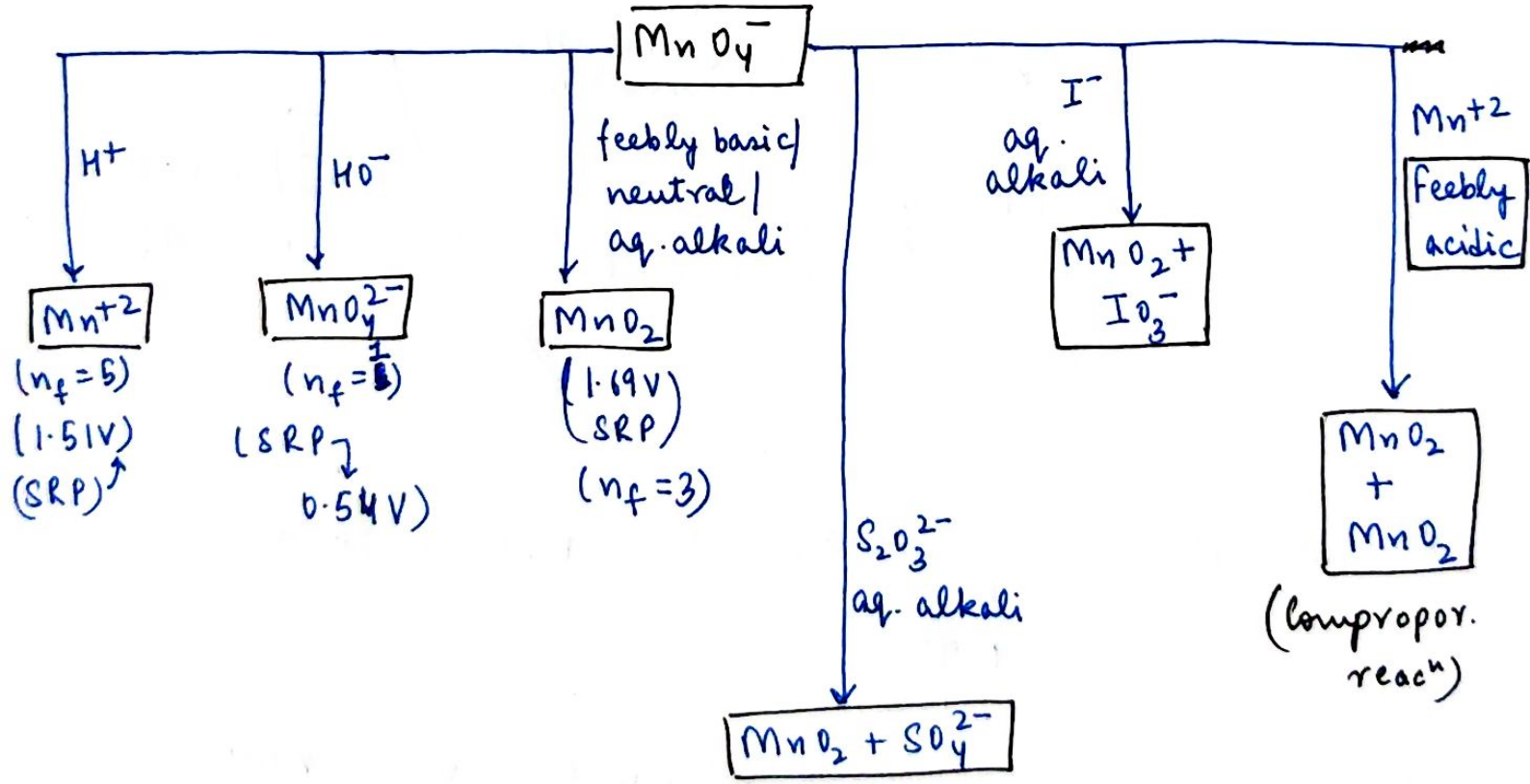
① $3MnO_4^{2-} + 4H^+ \xrightarrow[\text{(Disproportionation reaction)}]{\text{D.R}} 2MnO_4^- + MnO_2 + 2H_2O$

② $2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + O_2 + 2KOH$

#


MnO_2 , Mn_3O_4
Al
Al_2O_3 + Mn
CO
'C'+
Δ
CO + Mn
CO_2 + MnO
H_2SO_4
$MnSO_4$ + H_2O
electrolysis
Mn
Δ
Mn-Hg
Hg
cathode
C Anode

O. Agent →


MnO_4^-
H^+
HO^-
feebly basic/ neutral/ aq. alkali
I^- aq. alkali
Mn^{+2} Feebly acidic
Mn^{+2}
(n_f = 5)
(1.51V)
(SRP)
MnO_4^{2-}
(n_f = 1)
(SRP → 0.54V)
MnO_2
(1.69V)
(SRP)
(n_f = 3)
MnO_2 + IO_3^-
MnO_2 + MnO_2
(comproport. reacn)
$S_2O_3^{2-}$ aq. alkali
MnO_2 + SO_4^{2-}